BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

The vessels and the rules of flow, measuring a blood pressure, how arterioles and capillaries do their jobs, venous return, and the reflexes and hormones that hold pressure steady. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

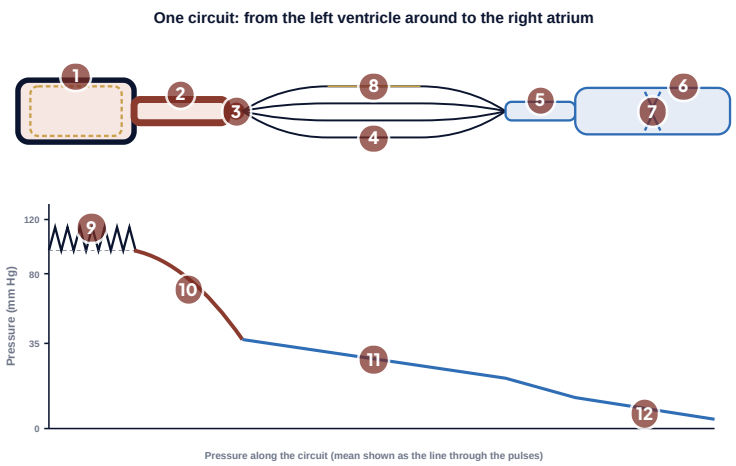
Name _____

A. The vessels, and the rules of flow

YOU SHOULD BE ABLE TO

Vessel structure and function. Match arteries, arterioles, capillaries, venules, and veins to their role in pressure buffering, resistance, exchange, and capacitance.

Pressure flow and resistance. Apply the relationship among flow, pressure gradient, and resistance and predict the effect of a change in vessel radius, blood viscosity, or vessel length on flow.



One circuit, with the pressure in each vessel

1. Elastic artery
2. Arteriole
3. Precapillary sphincter
4. Capillary
5. Venule
6. Vein
7. Venous valve
8. Endothelium
9. Pulse pressure in the artery
10. Pressure drop across the arterioles
11. Capillary pressure
12. Venous pressure

WHAT YOU ARE LOOKING AT

One lap of the systemic circuit, left to right: artery, arteriole, capillaries, venule, vein. The graph underneath is the pressure in each vessel. Wall thickness is drawn to tell you what each vessel is built to do.

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

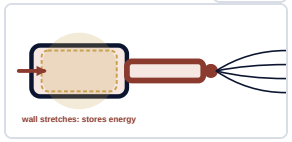
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Name _____

A. The vessels, and the rules of flow (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

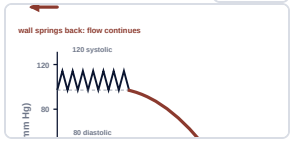
STEP 1



The ventricle ejects into an elastic artery

During systole the aorta's elastic wall stretches and stores some of the pulse's energy; pressure peaks near 120 mm Hg

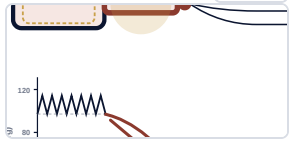
STEP 2



Diastole: the artery recoils

The stretched wall springs back and keeps pushing blood forward between beats, so flow never stops and pressure only falls to about 80; the pulses on the graph are this stretch and recoil

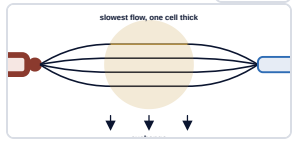
STEP 3



The arteriole resists

A thick ring of smooth muscle makes the arteriole the narrowest adjustable point; most of the pressure is spent getting through it (about 93 down to 35) and the pulse disappears

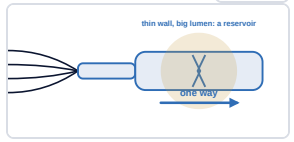
STEP 4



The capillary exchanges

Total cross-sectional area is enormous, so velocity is slowest here; with one cell of wall, gases, nutrients, waste and fluid cross

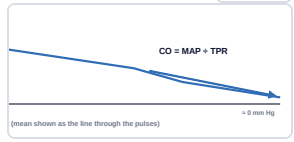
STEP 5



Venules and veins collect and store

Pressure is now a few mm Hg; the thin, compliant veins hold about 60% of the blood, and valves keep it moving one way

STEP 6



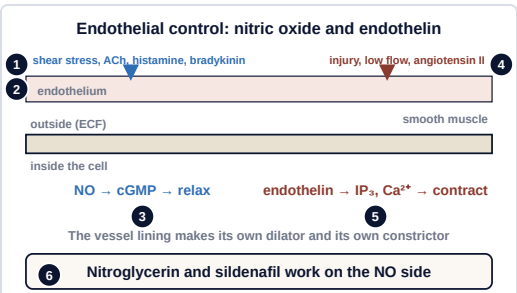
Back to the right atrium at about 0 mm Hg

The whole loop runs on one rule: flow = pressure gradient ÷ resistance, so cardiac output = MAP ÷ total peripheral resistance

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

Pressure falls all the way around, and most of it is spent at the arterioles. Resistance goes with 1 ÷ radius^4, so halving an arteriole's radius raises its resistance 16 times: small changes in arteriolar tone are the body's main way to steer flow and set blood pressure.

8 Endothelium: how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

1. A dilating signal reaches the lining: blood dragging across it (shear stress), or acetylcholine, histamine or bradykinin binding receptors on the endothelial cell. **2.** The endothelial cell switches on its nitric oxide synthase and makes nitric oxide, a gas that diffuses straight through membranes. **3.** Nitric oxide enters the smooth muscle cell underneath, raises cGMP, and the muscle relaxes: the vessel dilates. **4.** A constricting signal reaches the lining instead: injury, low flow, or angiotensin II. **5.** The endothelial cell makes endothelin, which binds the smooth muscle and works through IP_3 and Ca^{2+}: the vessel constricts. **6.** The lining makes its own dilator and its own constrictor. Nitroglycerin and sildenafil both act on the nitric oxide side. **What it does:** Makes nitric oxide and endothelin, controls what crosses, keeps blood from clotting on it **Why it matters:** Endothelial damage is the start of atherosclerosis

Extra notes and drawings

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A. The vessels, and the rules of flow (continued)

THE RULES OF FLOW

Flow = **pressure gradient ÷ resistance (F = ΔP ÷ R); for the whole circuit, CO = MAP ÷ TPR**

Resistance depends on **vessel length, blood viscosity, and radius; R is proportional to length × viscosity ÷ radius⁴**

Halve the radius and **resistance rises 16 times, so flow falls to one sixteenth at the same pressure**

Double the viscosity (polycythemia) and **resistance doubles, flow halves**

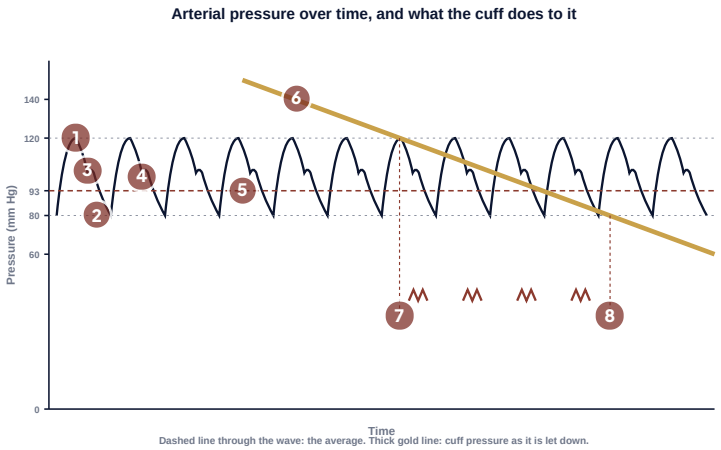
Length matters when **new vessels grow (obesity, a tumor); otherwise it is fixed**

Velocity of flow is **flow ÷ total cross-sectional area: fastest in the aorta, slowest in the capillaries, faster again in the veins**

B. Arterial blood pressure, and how you measure it

YOU SHOULD BE ABLE TO

Arterial blood pressure. Define systolic, diastolic, pulse, and mean arterial pressure, calculate mean arterial pressure, and explain how it is determined by cardiac output and total peripheral resistance.
Blood pressure measurement. Measure blood pressure by auscultation, explain the origin of the Korotkoff sounds, and record the response to a postural or exercise challenge.



The arterial pressure wave and the cuff

1. **Systolic pressure**
2. **Diastolic pressure**
3. **Dicrotic notch**
4. **Pulse pressure**
5. **Mean arterial pressure**
6. **Cuff pressure**
7. **First Korotkoff sound**
8. **Last Korotkoff sound**

WHAT YOU ARE LOOKING AT

Left: the pressure in a large artery, beat by beat. Right: the same wave with a cuff pressure (gold) being let down across it. The sounds you hear through the stethoscope are drawn below the wave.

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

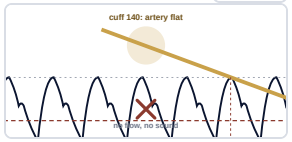
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B. Arterial blood pressure, and how you measure it (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

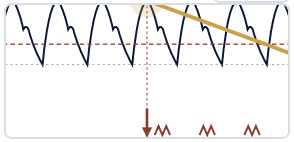
STEP 1



Cuff above systolic: the artery is shut

At 140 mm Hg the cuff squashes the brachial artery flat; no blood passes, no sound

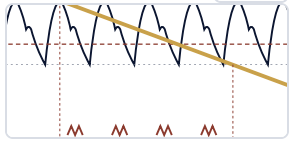
STEP 2



Cuff drops below systolic: the first tap

The first time the pulse peak exceeds cuff pressure, blood squirts through the narrowed artery in a turbulent burst; that tap is the first Korotkoff sound and the cuff reading is systolic

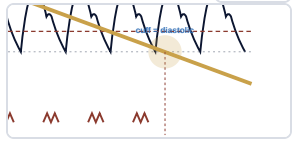
STEP 3



Sounds continue while the cuff is between the two

Each beat the artery opens briefly and snaps shut: tapping with every pulse

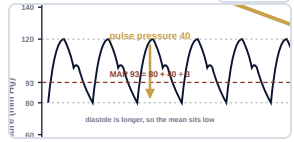
STEP 4



Cuff drops below diastolic: silence

The artery stays open all the time, flow is smooth (laminar) and makes no noise; the cuff reading when the sounds vanish is diastolic

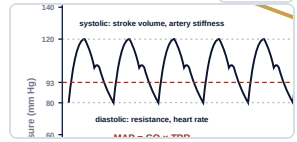
STEP 5



Pulse pressure and mean pressure

Pulse pressure = systolic − diastolic (120 − 80 = 40). Mean arterial pressure = diastolic + ⅓ pulse pressure (80 + 13 = 93), because diastole lasts longer than systole

STEP 6



What sets each number

Systolic rises with stroke volume and stiff arteries; diastolic rises with resistance (arteriolar tone) and heart rate; MAP = cardiac output × total peripheral resistance

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

Two numbers from a cuff tell you about the pump and the pipes: a high diastolic means the arterioles are tight, a wide pulse pressure means stiff arteries or a big stroke volume. MAP is the number the body defends, and it has only two levers: cardiac output and resistance.

WORK THE NUMBERS

Pulse pressure = systolic − diastolic **120 − 80 = 40 mm Hg**

MAP = diastolic + ⅓ pulse pressure **80 + 40 ÷ 3 = 93 mm Hg (closer to diastolic because diastole lasts longer)**

MAP = cardiac output × total peripheral resistance **raise either and MAP rises; the baroreflex adjusts both**

A patient at 150/70 **pulse pressure 80, MAP 97: stiff arteries (wide pulse pressure), MAP about normal**

A patient at 100/85 **pulse pressure 15, MAP 90: small stroke volume with tight arterioles, as in early shock**

Posture or exercise in lab **standing: a brief dip then recovery; exercise: systolic up, diastolic about the same or slightly down**

Extra notes and drawings

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Name _____

C. How an arteriole decides its own tone

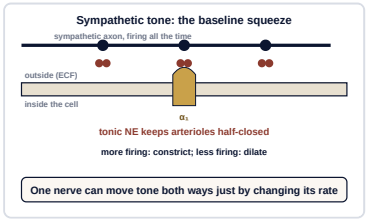
YOU SHOULD BE ABLE TO

Local control of blood flow. Explain myogenic, metabolic, and endothelial control of arte…se to increased tissue metabolism.

HOW AN ARTERIOLE DECIDES ITS OWN TONE

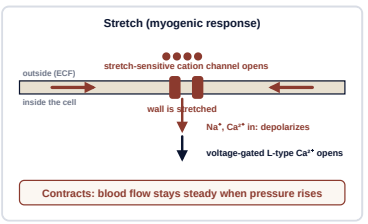
Arteriolar tone

Sympathetic tone



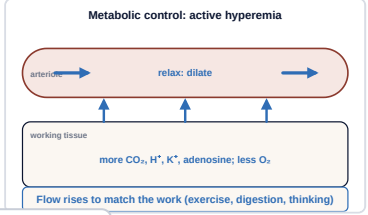
Tonic norepinephrine on α_1 receptors keeps most arterioles partly constricted; more firing constricts, less firing dilates

Myogenic



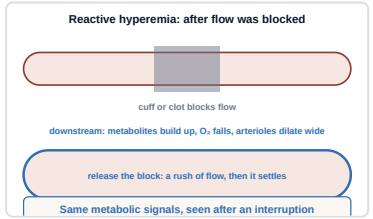
Stretch opens cation channels in the smooth muscle, it depolarizes and contracts; drop the pressure and it relaxes

Metabolic (active hyperemia)



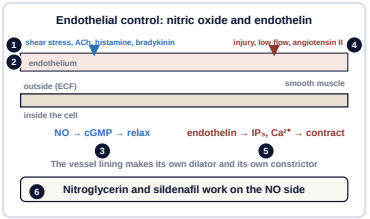
…sue makes more CO_2, H^+, K^+ …e and uses up O_2; all of them …riole

…yperemia



Block the flow, metabolites build up and the arterioles dilate; release the block and flow overshoots before settling

Endothelial



The lining makes nitric oxide (cGMP, relax) in response to shear stress, ACh, histamine and bradykinin, and endothelin (contract) in response to injury

Extra notes and drawings

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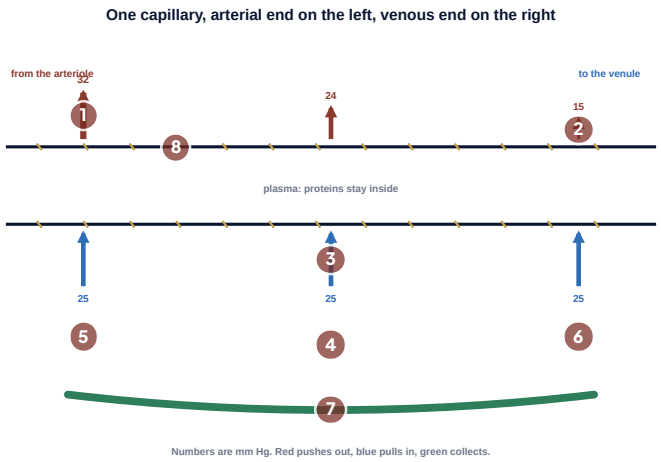
D. Capillary exchange, and edema

YOU SHOULD BE ABLE TO

Capillary exchange and Starling forces. Identify the four Starling forces at a capillary and determine the net direction of fluid movement at the arterial and venous ends.

Edema. Explain how increased capillary hydrostatic pressure, reduced plasma protein, increased permeability, or lymphatic blockage each produce edema and match a clinical scenario to its mechanism.

Lymphatic return. State the volume of fluid the lymphatic system returns each day and trace lymph from an interstitial space to the subclavian vein.



Starling forces along one capillary

1. Capillary hydrostatic pressure, arterial end
2. Capillary hydrostatic pressure, venous end
3. Plasma colloid osmotic pressure
4. Interstitial fluid
5. Net filtration (arterial end)
6. Net absorption (venous end)
7. Lymphatic capillary
8. Capillary wall (endothelium with pores)

WHAT YOU ARE LOOKING AT

One capillary opened out, blood flowing left to right. Red arrows: pressure pushing fluid out through the pores. Blue arrows: the pull of plasma proteins drawing fluid back in. Green: the lymphatic that collects what is left over.

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

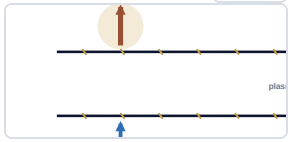
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Name _____

D. Capillary exchange, and edema (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

STEP 1



At the arterial end, hydrostatic pressure wins

Capillary hydrostatic pressure is about 32 mm Hg pushing out; plasma colloid osmotic pressure is about 25 pulling in; net 7 mm Hg outward: filtration

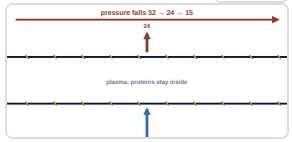
STEP 2



Fluid leaves, proteins stay

Water and small solutes pass through the pores between endothelial cells; albumin is too big, so the osmotic pull inside stays at 25

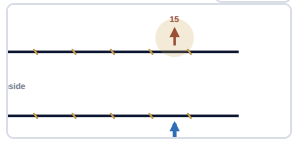
STEP 3



Pressure falls along the capillary

Friction drops hydrostatic pressure from 32 to about 15 by the venous end; the osmotic pull does not change

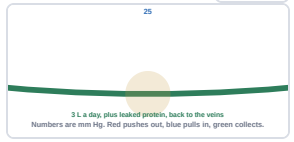
STEP 4



At the venous end, the osmotic pull wins

25 in against 15 out: net 10 mm Hg inward: absorption. Most of what left comes back

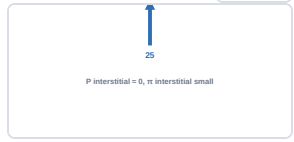
STEP 5



The lymphatic takes the remainder

About 3 L a day across the body is not reabsorbed; lymphatic capillaries pick it up, with any protein that leaked, and return it to the veins near the heart

STEP 6



The interstitial forces are small

Interstitial hydrostatic pressure is near zero and interstitial protein is low, so in health the balance is set by the two capillary forces

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

Filtration at the start, absorption at the end, lymph for the difference. Raise capillary pressure, lower plasma protein, open the pores, or block the lymphatics and fluid piles up in the tissue: that is edema, and each cause maps to a disease.

Extra notes and drawings

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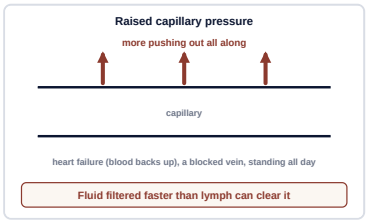
Name _____

D. Capillary exchange, and edema (continued)

FOUR WAYS TO GET EDEMA

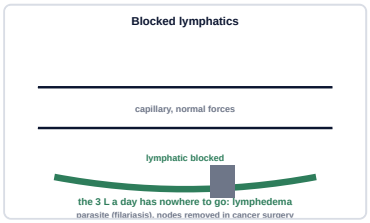
Edema

Raised capillary pressure



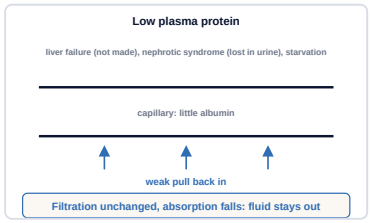
More hydrostatic push along the whole capillary: more filtration, less absorption

Blocked lymphatics



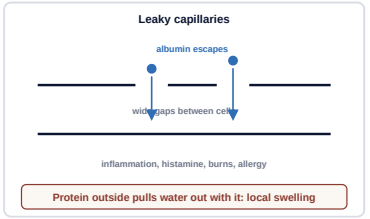
The 3 L a day that is not reabsorbed has nowhere to go

Low plasma protein



Less albumin means less osmotic pull back in; filtration continues, absorption fails

Leaky capillaries



Inflammation and histamine widen the pores; albumin escapes and takes its osmotic pull outside with it

LYMPH, BACK TO THE BLOOD: ABOUT 3 L A DAY

STEP 1
Interstitial space
Of the roughly 20 L filtered out of capillaries each day, about 17 L is reabsorbed at the venous end; the 3 L left behind, plus any protein that leaked, is lymph

STEP 2
Lymphatic capillary
Blind-ended tube whose endothelial cells overlap like one-way flaps: fluid pushes them open to get in and shuts them to get out

STEP 3
Lymphatic vessels
Valves and the skeletal muscle pump move the lymph along, the same way they move venous blood

STEP 4
Lymph nodes
Filtered through nodes on the way: immune cells inspect it

STEP 5
Thoracic duct, or right lymphatic duct
The thoracic duct collects from the whole body below the diaphragm and the left upper quarter; the right lymphatic duct takes the right upper quarter

STEP 6
Subclavian veins
Both ducts empty into the junction of the internal jugular and subclavian veins, so the 3 L is back in the blood, and the plasma protein with it

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

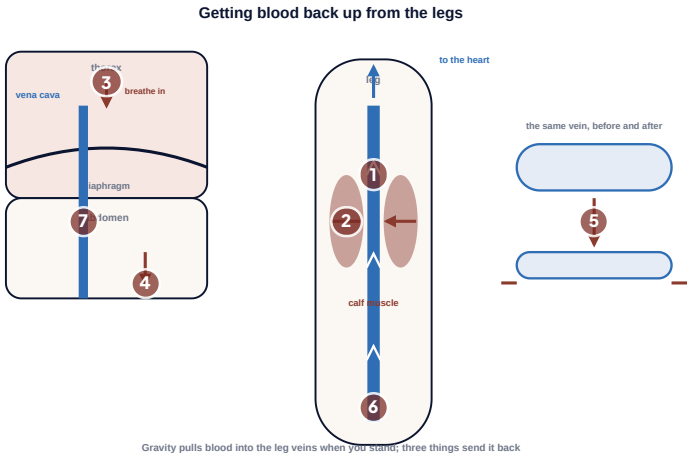
The vessels and the rules of flow, measuring a blood pressure, how arterioles and capillaries do their jobs, venous return, and the reflexes and hormones that hold pressure steady. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

E. Venous return

YOU SHOULD BE ABLE TO

Venous return. Explain the skeletal muscle pump, the respiratory pump, and venoconstriction and predict the effect of prolonged standing on venous return and stroke volume.



Three pumps that bring blood back

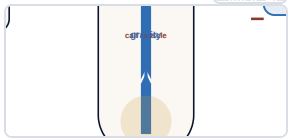
1. Venous valve
2. Skeletal muscle pump
3. Respiratory pump: thoracic pressure falls on inspiration
4. Respiratory pump: abdominal pressure rises
5. Sympathetic venoconstriction
6. Venous pooling in the legs
7. Vena cava

WHAT YOU ARE LOOKING AT

Left: the trunk with the diaphragm between thorax and abdomen and the vena cava running up through both. Middle: a leg vein with its valves and the calf muscle around it. Right: a vein before and after sympathetic constriction.

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

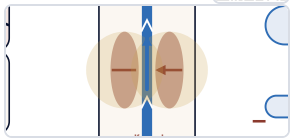
STEP 1



Standing: gravity fills the leg veins

Veins are compliant, so blood pools below the heart; venous return, end-diastolic volume and stroke volume all fall

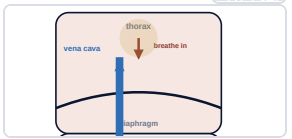
STEP 2



The skeletal muscle pump

Each contraction squeezes the deep veins; the valve above opens and the valve below closes, so blood can only go up

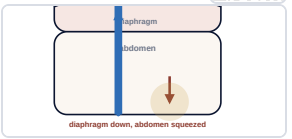
STEP 3



The respiratory pump, inhale

Thoracic pressure falls as the chest expands, which draws blood up the vena cava toward the right atrium

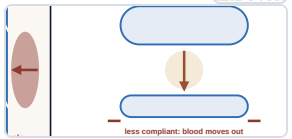
STEP 4



The respiratory pump, below the diaphragm

The descending diaphragm raises abdominal pressure and pushes abdominal venous blood toward the chest

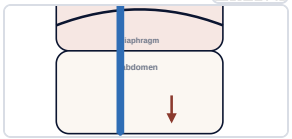
STEP 5



Sympathetic venoconstriction

Norepinephrine on α_1 receptors tightens the veins; the reservoir holds less, so blood moves to the heart

STEP 6



More venous return, more stroke volume

Venous return fills the ventricle; by Frank-Starling the next beat ejects more

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

Venous return is what preload is made of. Stand still for a long time and the pumps stop working: venous return, stroke volume and blood pressure fall and you can faint. Walk, breathe deeply, or let the sympathetic system tighten the veins, and the blood comes back.

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

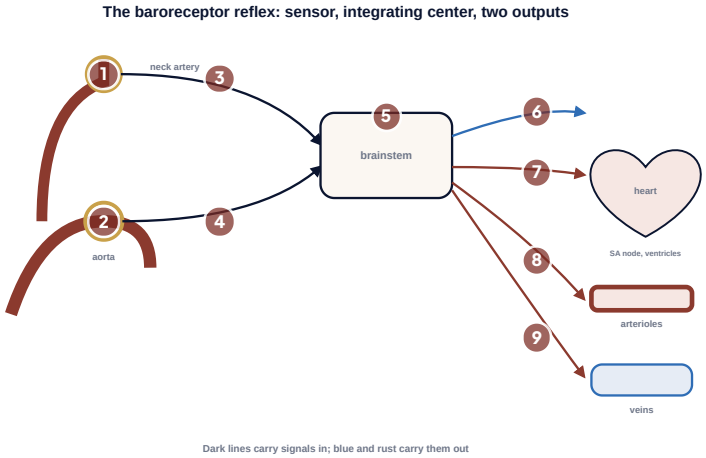
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Name _____

F. Holding pressure steady: the reflex and the hormones

YOU SHOULD BE ABLE TO

Baroreceptor reflex. Trace the baroreceptor reflex from the carotid sinus and aortic arch through the medulla to the heart and vessels and state the response to a fall in blood pressure.
Hormonal control of blood pressure. Describe the roles of angiotensin II, aldosterone, antidiuretic hormone, and atrial natriuretic peptide in long term blood pressure control.
Adrenal cortex. Match the three cortical zones to aldosterone, cortisol, and the adrenal androgens and state the primary action of each.



The baroreceptor reflex

1. Carotid sinus baroreceptor
2. Aortic arch baroreceptor
3. Glossopharyngeal nerve (CN IX), afferent
4. Vagus nerve (CN X), afferent
5. Medullary cardiovascular control center
6. Parasympathetic output to the SA node
7. Sympathetic output to the heart
8. Sympathetic output to arterioles
9. Sympathetic output to veins

WHAT YOU ARE LOOKING AT

Left: the two places pressure is sensed, the carotid sinus and the aortic arch. Middle: the medulla. Right: the three targets. Dark lines carry the signal in; blue is the parasympathetic output, rust the sympathetic output. Read it for a fall in pressure, such as standing up.

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

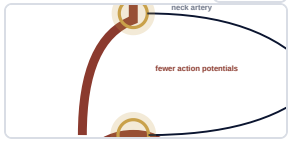
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F. Holding pressure steady: the reflex and the hormones (continued)

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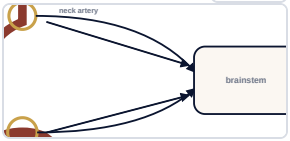
STEP 1



Pressure falls, the baroreceptors stretch less

Carotid sinus and aortic arch receptors fire slower

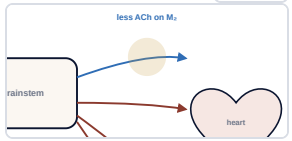
STEP 2



Fewer signals reach the medulla

Along the glossopharyngeal (carotid sinus) and vagus (aortic arch) nerves to the cardiovascular control center

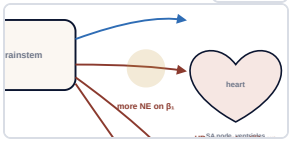
STEP 3



The medulla cuts the parasympathetic brake

Less vagal acetylcholine on M_2 receptors: the SA node speeds up

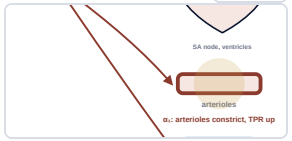
STEP 4



and raises sympathetic drive to the heart

Norepinephrine on β_1: heart rate and contractility rise, so cardiac output rises

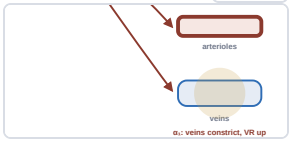
STEP 5



Sympathetic drive constricts the arterioles

Norepinephrine on α_1: total peripheral resistance rises

STEP 6



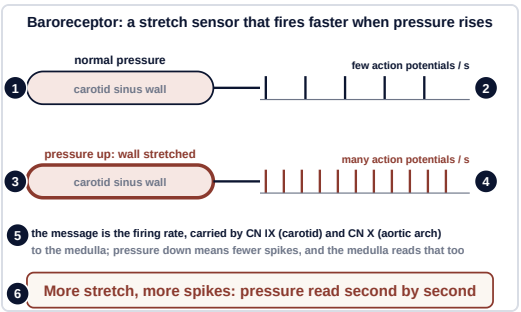
and the veins

Venoconstriction raises venous return and stroke volume

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

Within a few heartbeats cardiac output and resistance both rise and MAP comes back toward its set point: MAP = CO × TPR, and the reflex pushes both. It works the other way for a rise in pressure. It is fast but it does not hold a grudge: in chronic hypertension it resets to the new pressure within a day or two, so the long-term work belongs to the kidney and its hormones.

1 Carotid sinus baroreceptor: how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

1. At normal pressure the carotid sinus wall is stretched a little. **2.** The baroreceptor nerve endings in the wall fire a steady, modest train of action potentials up the glossopharyngeal nerve (from the aortic arch, the vagus). **3.** Pressure rises: the wall stretches more. **4.** The endings fire faster. More spikes per second is the message that pressure is up; fewer spikes is the message that it is down. **5.** The medulla reads the firing rate and adjusts sympathetic and parasympathetic output within a few seconds. **6.** More stretch, more spikes: that is the whole sensor. **What it does:** Fire faster when stretched by higher pressure, slower when pressure falls **Why it matters:** Guards pressure to the brain; pressing on the neck here can slow the heart

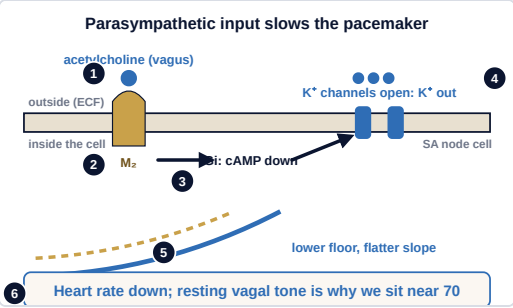
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F. Holding pressure steady: the reflex and the hormones (continued)

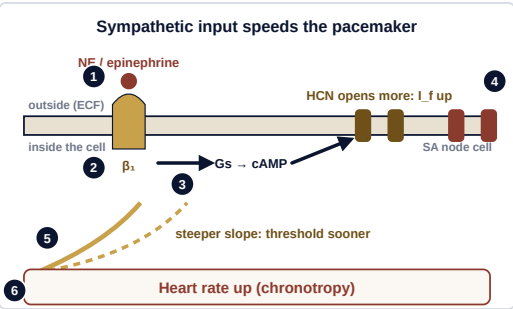
6 Parasympathetic output to the SA node: how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

1. Acetylcholine from the vagus nerve arrives at the SA node cell. **2.** It binds the M_2 receptor, a G protein-coupled receptor on the membrane. **3.** M_2 works through Gi: adenylyl cyclase is inhibited and cAMP falls, so the funny channels open less and the drift is flatter. **4.** Gi also opens K^+ channels directly; K^+ leaves and the membrane sits lower (hyperpolarized), so the floor drops. **5.** Flatter slope from a lower floor: threshold takes longer to reach. **6.** Heart rate goes down. The vagus fires all the time at rest, which is why the resting rate sits near 70 instead of the node's own 100. **What it does:** Slows the pacemaker **Why it matters:** This is the fast brake on heart rate, and resting vagal tone is why the resting rate is near 70 rather than 100

7 Sympathetic output to the heart: how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

1. Norepinephrine from the sympathetic nerve, or epinephrine from the adrenal medulla, arrives at the SA node cell. **2.** It binds the β_1 receptor, a G protein-coupled receptor on the membrane. **3.** β_1 works through Gs: adenylyl cyclase is switched on and cAMP rises inside the cell. **4.** cAMP binds the funny (HCN) channel directly and opens it more; I_f is larger. cAMP also activates PKA, which opens more L-type Ca^{2+} channels (the red channel). **5.** The pacemaker potential climbs more steeply and reaches threshold sooner. **6.** Heart rate goes up. In ventricular muscle the same cAMP raises contractility, so each beat is stronger too. **What it does:** Raises heart rate and contractility **Why it matters:** Raises cardiac output within seconds

Extra notes and drawings

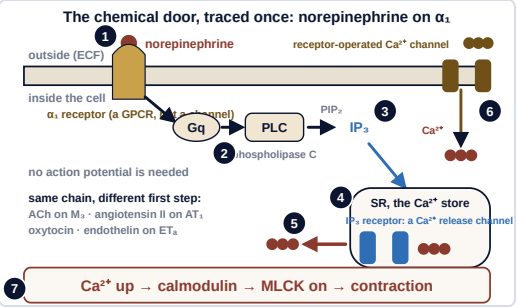
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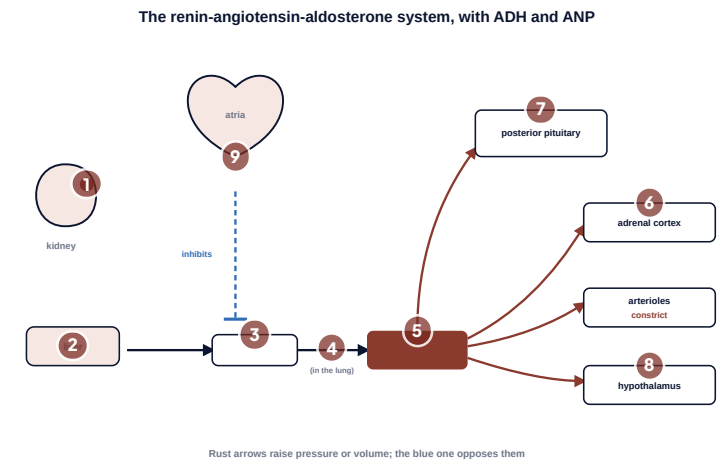
F. Holding pressure steady: the reflex and the hormones (continued)

8 Sympathetic output to arterioles: how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

1. Norepinephrine lands on the α_1 receptor on the outside of the smooth muscle cell. The receptor is a G protein–coupled receptor: nothing passes through it. **2.** The receptor switches on its G protein, Gq, on the inside of the membrane. **3.** Gq switches on phospholipase C, which cuts a membrane lipid (PIP_2) and releases IP_3 into the cytoplasm. **4.** IP_3 floats to the sarcoplasmic reticulum and opens its Ca^{2+} release channel, the IP_3 receptor. **5.** Ca^{2+} pours out of the store into the cytoplasm. **6.** At the same time, the same signal opens the receptor-operated Ca^{2+} channel in the membrane, and more Ca^{2+} comes in from outside. No action potential anywhere in this chain. **7.** Ca^{2+} binds calmodulin, calmodulin switches on MLCK, MLCK phosphorylates myosin, and the cell contracts. Acetylcholine on M_3, angiotensin II on AT_1, oxytocin and endothelin run the same chain from step 1 with their own receptor. **What it does:** Constricts arterioles, raising total peripheral resistance **Why it matters:** Resistance is one of the two terms in MAP = CO × TPR, so this output moves blood pressure quickly



The renin-angiotensin-aldosterone system, with ADH and ANP

1. Juxtaglomerular cells (renin)
2. Angiotensinogen (from the liver)
3. Angiotensin I
4. Angiotensin-converting enzyme (ACE)
5. Angiotensin II
6. Aldosterone (adrenal cortex)
7. Antidiuretic hormone (ADH, vasopressin)
8. Thirst center
9. Atrial natriuretic peptide (ANP)

WHAT YOU ARE LOOKING AT

The long-term lever on blood pressure: a chain of hormones that ends in the kidney keeping salt and water. Follow the arrows from the kidney at the left, through the blood, to the targets on the right. ANP, in blue, pulls the other way.

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

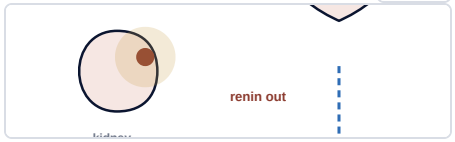
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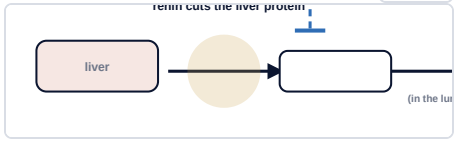
STEP 1



The kidney releases renin

Juxtaglomerular cells release renin when renal perfusion pressure falls, when the macula densa senses less NaCl, or when sympathetic nerves fire on β_1 receptors

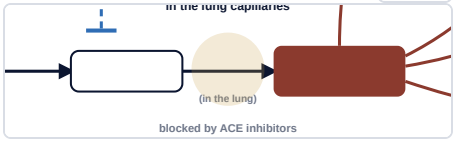
STEP 2



Renin cuts angiotensinogen

A liver protein always present in plasma becomes angiotensin I, which is inactive

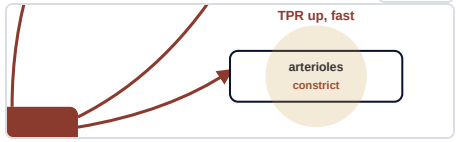
STEP 3



ACE makes angiotensin II

Angiotensin-converting enzyme on the lung endothelium converts I to II; this is the step the -pril drugs block

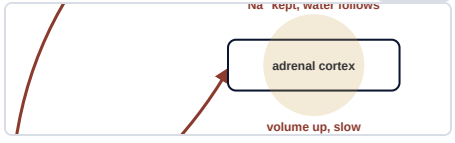
STEP 4



Angiotensin II constricts arterioles

Resistance rises at once: a fast effect

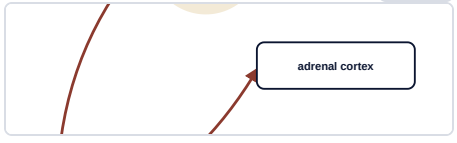
STEP 5



Angiotensin II triggers aldosterone

The zona glomerulosa releases aldosterone; the distal nephron reabsorbs Na^+ and water follows: more volume over hours to days

STEP 6



Angiotensin II triggers ADH and thirst

The posterior pituitary releases ADH (water reabsorbed in the collecting duct) and the hypothalamus makes you drink

STEP 7



ANP opposes it

When the atria are overstretched they release atrial natriuretic peptide: the kidney excretes Na^+ and water, arterioles dilate, and renin, aldosterone and ADH are all inhibited

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

The baroreflex fixes pressure in seconds by changing resistance and output; this system fixes it over hours and days by changing blood volume. Most blood pressure drugs act here: ACE inhibitors, angiotensin receptor blockers, aldosterone antagonists, and diuretics that undo the volume.

Extra notes and drawings

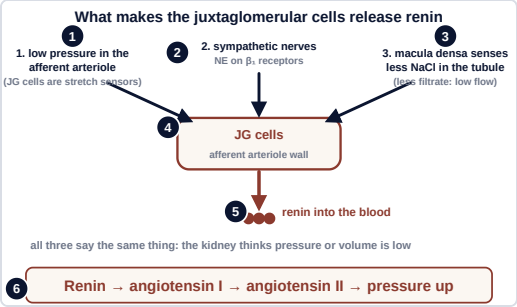
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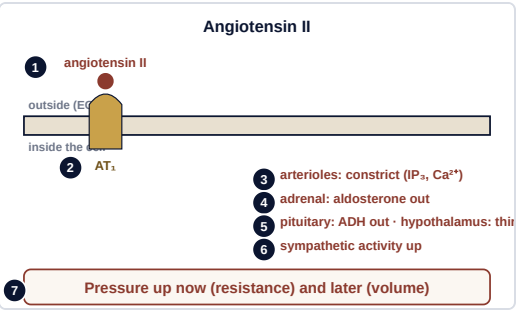
1 Juxtaglomerular cells (renin): how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

1. Pressure in the afferent arteriole falls. The juxtaglomerular cells sit in its wall and are stretch sensors, so less stretch is their first signal. **2.** Sympathetic nerves release norepinephrine onto β_1 receptors on the JG cells. **3.** The macula densa, next door in the distal tubule, senses less NaCl going past (less filtrate) and signals the JG cells. **4.** All three signals mean the same thing to the kidney: pressure or volume is low. **5.** The JG cells release renin into the blood. **6.** Renin starts the cascade: angiotensinogen to angiotensin I, ACE turns that into angiotensin II, and pressure comes up. **What it does:** Release renin when renal pressure falls, when the macula densa sees less NaCl, or when sympathetic β_1 receptors are stimulated **Why it matters:** The kidney's own pressure sensor starts the cascade

5 Angiotensin II: how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

1. Angiotensin II arrives in the blood, made from angiotensin I by ACE in the lungs after renin started the cascade. **2.** It binds the AT_1 receptor, a Gq-coupled receptor, on each of its targets. **3.** On arterioles: IP_3 and Ca^{2+} rise and the smooth muscle constricts. Resistance, and so pressure, rise within seconds. **4.** On the adrenal cortex: aldosterone is released, which will raise blood volume over days. **5.** On the posterior pituitary: ADH is released; in the hypothalamus: thirst. Both add water. **6.** On sympathetic nerves: activity goes up, adding to the constriction. **7.** One hormone, one direction: pressure up now by resistance, and later by volume. **What it does:** Constricts arterioles, triggers aldosterone and ADH, drives thirst, and raises sympathetic activity **Why it matters:** It raises pressure right away by raising resistance, and over the following days by raising blood volume

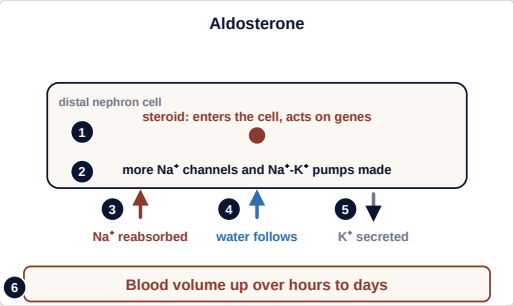
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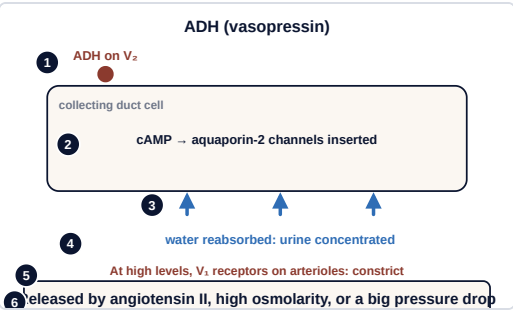
6 Aldosterone (adrenal cortex): how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

1. Aldosterone, a steroid from the zona glomerulosa of the adrenal cortex, passes straight through the membrane of a distal nephron cell. **2.** Inside, it binds a receptor that acts on genes: the cell makes more Na^+ channels and more Na^+-K^+ pumps. Because it works through new proteins it is slow, hours to days. **3.** More Na^+ is reabsorbed from the tubule back into the blood. **4.** Water follows the sodium, so blood volume rises. **5.** K^+ is secreted into the tubule in exchange, which is why aldosterone excess lowers blood potassium. **6.** Blood volume, and with it pressure, rise over hours to days: the slow lever. **What it does:** Makes the distal nephron reabsorb Na^+ (and water follows) and secrete K^+ **Why it matters:** The slow lever: more volume, more pressure, over hours to days

7 Antidiuretic hormone (ADH, vasopressin): how it works



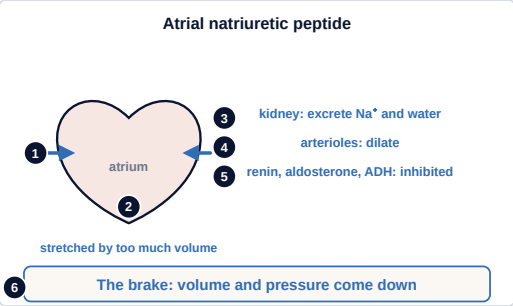
Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

1. ADH (vasopressin) from the posterior pituitary binds the V_2 receptor on a collecting duct cell. **2.** V_2 raises cAMP inside the cell, and the cell inserts aquaporin-2 water channels into its tubule-facing membrane. **3.** Water leaves the tubule through the channels and returns to the blood. **4.** The urine becomes concentrated and blood volume is protected. **5.** At high levels ADH also binds V_1 receptors on arterioles and constricts them, which is why it has the second name vasopressin. **6.** It is released by angiotensin II, by a rise in plasma osmolarity, and by a large fall in pressure. **What it does:** Makes the collecting duct reabsorb water; at high levels constricts arterioles **Why it matters:** Released by angiotensin II, by high osmolarity, and by a large fall in pressure

Extra notes and drawings

F. Holding pressure steady: the reflex and the hormones (continued)

9 Atrial natriuretic peptide (ANP): how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

1. Too much blood volume stretches the walls of the atria. **2.** The stretched atrial muscle cells release atrial natriuretic peptide into the blood. **3.** At the kidney, ANP makes the nephron excrete more Na^+ and water. **4.** On arterioles, ANP relaxes the smooth muscle: they dilate and resistance falls. **5.** ANP inhibits renin, aldosterone and ADH, so the whole volume-raising cascade is switched off. **6.** Volume and pressure come down. This is the only one of the hormones that pushes pressure down, and it is released only when there is already too much. **What it does:** Makes the kidney excrete Na^+ and water, dilates arterioles, and inhibits renin, aldosterone and ADH **Why it matters:** It is the brake on the whole system, released when there is already too much volume

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

The vessels and the rules of flow, measuring a blood pressure, how arterioles and capillaries do their jobs, venous return, and the reflexes and hormones that hold pressure steady. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

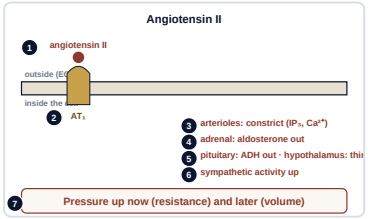
Name _____

F. Holding pressure steady: the reflex and the hormones (continu

HORMONES THAT HOLD PRESSURE OVER DAYS

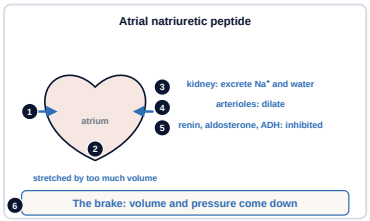
Blood volume

Angiotensin II



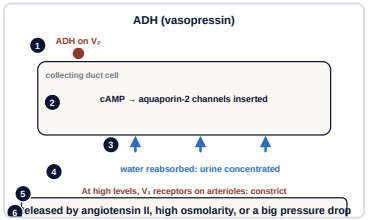
Constricts arterioles now; triggers aldosterone, ADH and thirst for later; raises sympathetic activity

Atrial natriuretic peptide



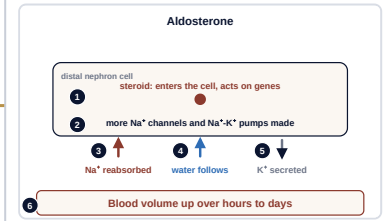
From stretched atria: excrete Na^+ and water, dilate arterioles, inhibit renin, aldosterone and ADH

ADH (vasopressin)



From the posterior pituitary: aquaporins in the collecting duct reabsorb water; at high levels constricts arterioles

Aldosterone



Steroid from the zona glomerulosa: the distal nephron reabsorbs Na^+ (water follows) and secretes K^+

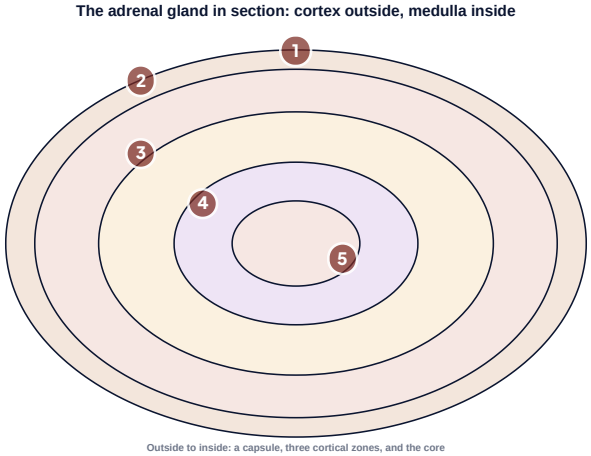
Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

The vessels and the rules of flow, measuring a blood pressure, how arterioles and capillaries do their jobs, venous return, and the reflexes and hormones that hold pressure steady. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

F. Holding pressure steady: the reflex and the hormones (continued)



The adrenal cortex in section

1. **Capsule**
2. **Zona glomerulosa**
3. **Zona fasciculata**
4. **Zona reticularis**
5. **Adrenal medulla**

THE ADRENAL CORTEX: ZONE, HORMONE, ACTION

	ZONA GLOMERULOSA	ZONA FASCICULATA	ZONA RETICULARIS
Hormone	aldosterone (mineralocorticoid)	cortisol (glucocorticoid)	adrenal androgens (DHEA)
Controlled by	angiotensin II and plasma K^+	ACTH from the anterior pituitary	ACTH
Primary action	kidney keeps Na^+ and water, secretes K^+: blood volume and pressure	raises blood glucose, breaks down protein and fat, suppresses inflammation and immunity: the stress hormone	weak androgens; pubic and axillary hair at puberty, libido in women
Too much	hyperaldosteronism: hypertension, low K^+	Cushing syndrome: central obesity, high glucose, thin skin	virilization

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

The vessels and the rules of flow, measuring a blood pressure, how arterioles and capillaries do their jobs, venous return, and the reflexes and hormones that hold pressure steady. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

G. Two stress tests: exercise, and losing blood

YOU SHOULD BE ABLE TO

Cardiovascular response to exercise. Predict the change in heart rate, stroke volume, cardiac output, total peripheral resistance, and regional blood flow distribution during moderate exercise.

Compensation in hemorrhage and shock. Sequence the compensatory responses to blood loss and explain the point at which compensation fails.

MODERATE EXERCISE: WHAT CHANGES

	CHANGE	WHY
Heart rate	up, to 150 or more	parasympathetic withdrawn, then sympathetic drive
Stroke volume	up	venous return up (muscle and respiratory pumps, venoconstriction) and contractility up
Cardiac output	up 4 to 5 times, to 20 to 25 L/min	HR × SV, both up
Total peripheral resistance	down	active hyperemia dilates muscle arterioles, which outweighs constriction elsewhere
Mean arterial pressure	up a little; systolic up, diastolic about the same	CO up more than TPR falls
Regional flow	muscle up 10 times, skin up (heat), heart up; gut and kidney down; brain unchanged	local dilation where it is needed, sympathetic constriction where it is not

LOSING BLOOD: COMPENSATION, THEN FAILURE

STEP 1
Blood volume falls
Less venous return, less EDV, less stroke volume, cardiac output and MAP fall

STEP 2
The baroreflex fires within seconds
Heart rate and contractility up (β_1), arterioles constrict in skin, gut and kidney (α_1), veins constrict: pale, cool skin and a fast weak pulse; flow is kept for brain and heart

STEP 3
Capillaries pull fluid back in
Lower capillary pressure means absorption beats filtration: interstitial fluid refills the plasma over minutes to hours (autotransfusion)

STEP 4
The hormones take over
Renin, angiotensin II, aldosterone and ADH keep Na^+ and water; thirst rises; the kidney makes little urine

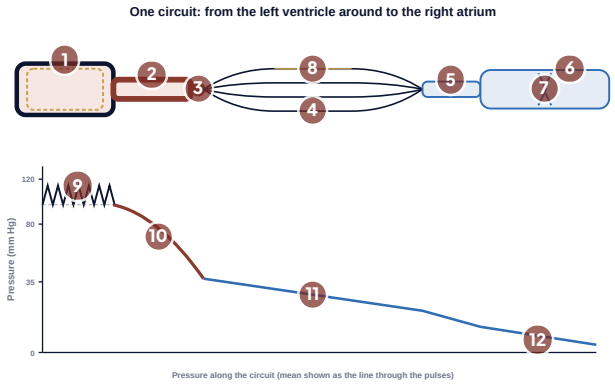
STEP 5
The slow repairs
Erythropoietin restores red cells over days; the liver replaces plasma proteins

STEP 6
Where compensation fails
Past roughly 30 to 40% loss, MAP can no longer be held: the heart and the vasomotor center themselves are underperfused, acidosis relaxes the arterioles, capillaries leak, and the fall in pressure feeds itself (progressive, then irreversible shock)

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

A. The vessels, and the rules of flow · Competency Vessel structure and function · Pressure flow and resistance · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____



One circuit, with the pressure in each vessel

1 Elastic artery

What it is Aorta and its big branches: thick wall with layers of elastin

What it does Stretches during ejection and recoils during diastole, so flow continues between beats

Why it matters Pressure reservoir: stiff arteries (age, atherosclerosis) raise systolic pressure and pulse pressure

2 Arteriole

What it is Small artery with a thick ring of smooth muscle and little elastin

What it does The main site of resistance; its radius sets how much flow a tissue gets and how much pressure is left for the capillaries

Why it matters Total peripheral resistance lives here; this is where blood pressure drugs act

3 Precapillary sphincter

What it is A ring of smooth muscle where a capillary leaves a metarteriole

What it does Opens or closes individual capillary beds

Why it matters Lets a tissue route blood to the cells that need it

6 Vein

What it is Thin wall, little muscle, large lumen, very compliant

What it does Capacitance vessel: holds about 60% of the blood at rest and gives it up when the sympathetic system tightens it

Why it matters The venous reservoir is what venous return and preload are drawn from

7 Venous valve

What it is Flaps of endothelium in limb veins

What it does Let blood flow toward the heart only

Why it matters Make the skeletal muscle pump work; failed valves give varicose veins

8 Endothelium

What it is The single cell layer lining every vessel

What it does Makes nitric oxide and endothelin, controls what crosses, keeps blood from clotting on it

Why it matters Endothelial damage is the start of atherosclerosis

9 Pulse pressure in the artery

What it is The swing between systolic and diastolic in the arteries

What it does Shows the stroke volume hitting an elastic wall

Why it matters The pulse you feel at the wrist is this pressure wave, not the blood itself

10 Pressure drop across the arterioles

What it is The steepest fall on the profile, from about 93 to about 35 mm Hg

What it does Resistance converts pressure into a controlled trickle for the capillaries

Why it matters Dilate the arterioles and pressure downstream rises; constrict them and it falls

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

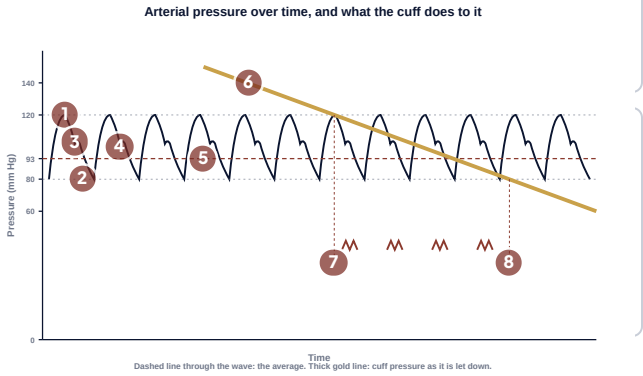
B. Arterial blood pressure, and how you measure it · Competency Arterial blood pressure · Blood pressure measurement · one card per numbered part, the same three lines: the slide reveals. Write the name, then what it is, what it does, and why it matters, as we go.

Name _____

What it is One layer of endothelium, 5 to 10 µm across, with a basement membrane

What it does The only vessel that exchanges gases, nutrients, waste and fluid across here

Why it matters This is where the body's tissues actually get what they need



The arterial pressure wave and the cuff

1 Systolic pressure

What it is The peak, about 120 mm Hg, during ventricular ejection

What it does Set by stroke volume and how stiff the arteries are

Why it matters This is the top number of a blood pressure reading

2 Diastolic pressure

What it is The trough, about 80 mm Hg, just before the next ejection

What it does Set mainly by total peripheral resistance and how long diastole lasts

Why it matters This is the bottom number of a blood pressure reading, and it tells you how tight the arterioles are

3 Dicrotic notch

What it is The small blip on the falling side of each pulse

What it does Marks the aortic valve closing (S2) and the brief backflow against it

Why it matters Separates ejection from diastole on a pressure trace

4 Capillary pressure

What it is About 32 mm Hg at the arterial end falling to about 15 at the venous end

What it does Pushes fluid out, through the pores (filtration)

Why it matters This is the pushing force in the Starling balance

5 Central venous pressure

What it is The pressure in the right atrium, near zero

What it does Rises when the heart cannot pump what returns to it

Why it matters Central venous pressure is a bedside measure of volume

6 Cuff pressure

What it is The pressure in the inflated cuff, let down slowly

What it does Above systolic it flattens the artery; below diastolic it does nothing

Why it matters The sounds you hear only exist while cuff pressure is between the two

7 First Korotkoff sound

What it is The first tapping heard as the cuff is let down

What it does Cuff pressure has just dropped below systolic: blood spurts through the squeezed artery in turbulent bursts

Why it matters The cuff reading at that moment is the systolic pressure

8 Last Korotkoff sound

What it is The sounds muffle and then stop

What it does Cuff pressure has dropped below diastolic: the artery stays open and flow is smooth and silent

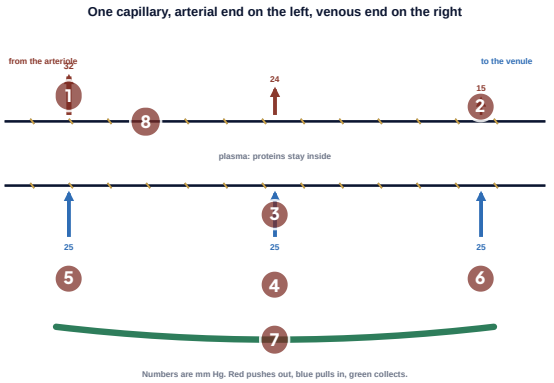
Why it matters The cuff reading at that moment is the diastolic pressure

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

Dr. Capillary exchange, and edema. Competency Capillary exchange and Starling forces · Edema · Lymphatic return · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____

4 Pulse pressure
What it is Systolic minus diastolic; about 40 mm Hg
What it does Rises with a bigger stroke volume or stiffer arteries
Why it matters A wide pulse pressure in an older adult means stiff arteries



Starling forces along one capillary

1 Capillary hydrostatic pressure, arterial end

What it is Blood pressure inside the capillary at its start, about 32 mm Hg
What it does Pushes fluid out through the pores
Why it matters The force that wins at the arterial end

2 Capillary hydrostatic pressure, venous end

What it is Blood pressure at the capillary's end, about 15 mm Hg
What it does Still pushes out, but weakly
Why it matters At this end the osmotic pull is stronger than the push, so fluid comes back in

3 Plasma colloid osmotic pressure

What it is About 25 mm Hg of pull from plasma proteins, mostly albumin, that cannot leave the capillary
What it does Draws water back in along the whole length
Why it matters Low albumin (liver failure, kidney loss, starvation) means fluid stays out: edema

6 Net absorption (venous end)

What it is In minus out: about 25 – 15 = 10 mm Hg inward
What it does Most of the filtered fluid returns here
Why it matters Across the body about 3 L a day is left over, not returned

7 Lymphatic capillary

What it is Blind-ended vessel with overlapping flaps that open inward
What it does Picks up the leftover 3 L a day and any protein that leaked, and returns it to the veins
Why it matters Block it and the limb swells (lymphedema)

8 Capillary wall (endothelium with pores)

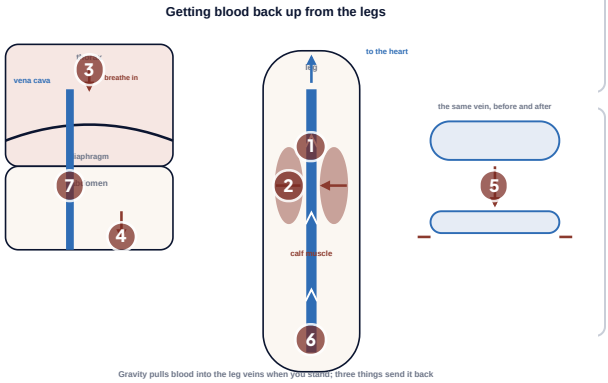
What it is One cell thick, with pores between the cells (and fenestrations in kidney and gut)
What it does Water and small solutes pass; proteins mostly do not
Why it matters Inflammation widens the pores, proteins leak, and the osmotic pull collapses

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

E. Venous return · Competency venous return · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does, and why it matters, as we go.

Name _____

4 Interstitial fluid
What it is The fluid between the cells, about a quarter of the extracellular fluid
What it does Is low in hydrostatic pressure near zero, and its protein is low, so its forces are small



Three pumps that bring blood back

1 Venous valve

What it is Flaps of endothelium in limb veins
What it does Let blood flow toward the heart only
Why it matters Make the skeletal muscle pump work; failed valves give varicose veins

2 Skeletal muscle pump

What it is Leg muscles squeezing the deep veins each time they contract
What it does Pushes blood toward the heart; the valves stop it falling back
Why it matters Standing still for a long time loses this pump and blood pools

3 Respiratory pump: thoracic pressure falls on inspiration

What it is The chest cavity's pressure drops as the ribs lift and the diaphragm descends
What it does Draws blood up the vena cava toward the right atrium
Why it matters Deep breathing raises venous return, which is one reason it helps when you feel faint

6 Venous pooling in the legs

What it is Blood settling in the compliant leg veins under gravity
What it does Lowers venous return, end-diastolic volume and stroke volume
Why it matters Why soldiers standing at attention faint, and why you feel faint standing up fast

7 Vena cava

What it is The two great veins returning blood to the right atrium
What it does Carry the venous return, at a pressure of only a few mm Hg
Why it matters Easily compressed; late pregnancy lying flat can squeeze it

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

F. Holding pressure steady: the reflex and the hormones · Competency Baroreceptor reflex · Hormonal control of blood pressure · Adrenal cortex · one card per numbered part, the same three lines the slide reveals · page 1 of 2. Write the name, then what it is, what it does and why it matters, as we go.

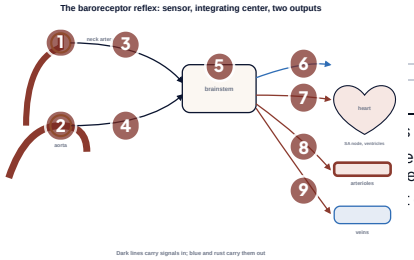
Name _____

9 Respiratory pump (abdominal pressure)

What it is The descending diaphragm squeezes the abdomen

What it does Pushes blood out of the abdominal veins toward the chest

Why it matters The two halves of the respiratory pump work together

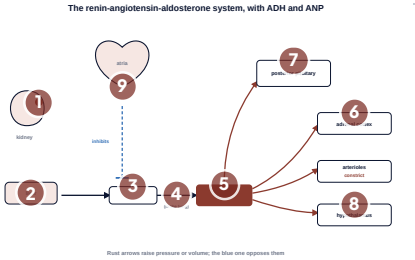


The baroreceptor reflex

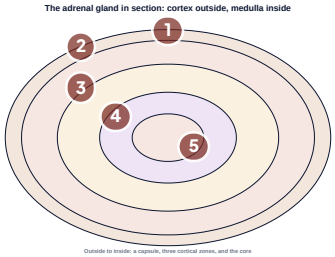
; smooth muscle

e same volume sits at
e heart

: of the venous reservoir



The renin-angiotensin-aldosterone system, with ADH and ANP



The adrenal cortex in section

1 Carotid sinus baroreceptor

What it is Stretch receptors in the wall of the carotid sinus, where the common carotid splits

What it does Fire faster when stretched by higher pressure, slower when pressure falls

Why it matters Guards pressure to the brain; pressing on the neck here can slow the heart

3 Glossopharyngeal nerve (CN IX), afferent

What it is The nerve carrying carotid sinus signals to the medulla

What it does It carries sensory signals in; it is the afferent arm of the reflex for the carotid sinus

Why it matters Cut it and the medulla no longer hears from the carotid sinus

4 Vagus nerve (CN X), afferent

What it is Carries aortic arch signals to the medulla

What it does Sensory input (the vagus also carries the parasympathetic output back out)

Why it matters The same nerve carries the sensory signal in and the parasympathetic signal back out

5 Medullary cardiovascular control center

What it is The nucleus of the solitary tract and the neighboring centers in the medulla

What it does Compares baroreceptor input with the set point and adjusts the two autonomic outputs

Why it matters This is the integrating center of the reflex, where the comparison with the set point happens

6 Parasympathetic output to the SA node

What it is Vagal fibers releasing acetylcholine on M_2 receptors

What it does Slows the pacemaker

Why it matters This is the fast brake on heart rate, and resting vagal tone is why the resting rate is near 70 rather than 100

7 Sympathetic output to the heart

What it is Sympathetic fibers releasing norepinephrine on β_1 receptors in the SA node and ventricles

What it does Raises heart rate and contractility

Why it matters Raises cardiac output within seconds

2 Angiotensinogen (from the liver)

What it is A plasma protein always present in blood

What it does Renin cuts it to angiotensin I

Why it matters Nothing happens without the enzyme; the substrate is just waiting

3 Angiotensin I

What it is A ten amino acid peptide, inactive

What it does Nothing yet; it circulates until an enzyme converts it

Why it matters It is a step with no action of its own, which is why the next enzyme matters so much

4 Angiotensin-converting enzyme (ACE)

What it is An enzyme on the endothelium, especially in the lungs

What it does Converts angiotensin I to angiotensin II

Why it matters ACE inhibitors (the -pril drugs) block this step to lower blood pressure

5 Angiotensin II

What it is The active hormone of the whole system

What it does Constricts arterioles, triggers aldosterone and ADH, drives thirst, and raises sympathetic activity

Why it matters It raises pressure right away by raising resistance, and over the following days by raising blood volume

6 Aldosterone (adrenal cortex)

What it is A steroid from the zona glomerulosa

What it does Makes the distal nephron reabsorb Na^+ (and water follows) and secrete K^+

Why it matters The slow lever: more volume, more pressure, over hours to days

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

F. Holding pressure steady: the reflex and the hormones · Competency Baroreceptor reflex · Hormonal control of blood pressure · Adrenal cortex · one card per numbered part, the same three lines the slide reveals · page 2 of 2 Write the name, then what it is, what it does and why it matters, as we go

2 Aortic arch baroreceptors
What it is Stretch receptors in the wall of the aortic arch
What it does Same signal as the carotid sinus, for the whole body
Why it matters Both sets report continuously; the reflex is tonic, not just emergency

1 Capsule
What it is The connective tissue coat of the gland
What it does Holds the gland together
Why it matters The zona glomerulosa sits right under it, which is how the zone got its name as the outer layer

2 Zona glomerulosa
What it is Outer cortical zone, cells in small clusters
What it does Makes aldosterone (a mineralocorticoid) under the control of angiotensin II and plasma K^+
Why it matters Because it controls salt, it controls blood volume, and so blood pressure

3 Zona fasciculata
What it is Middle and thickest zone, cells in straight columns
What it does Makes cortisol (a glucocorticoid) under ACTH
Why it matters Stress, blood glucose, and the immune brake; too much is Cushing syndrome

4 Zona reticularis
What it is Inner cortical zone, cells in a net
What it does Makes the adrenal androgens (DHEA) under ACTH
Why it matters These androgens drive pubic and axillary hair at puberty and make a small contribution in adults

5 Adrenal medulla
What it is The core: chromaffin cells, a modified sympathetic ganglion
What it does Releases epinephrine and norepinephrine into blood
Why it matters This is the fight-or-flight hormone source from last week

6 Sympathetic output to arterioles
What it is Norepinephrine on α_1 receptors in arteriolar smooth muscle
What it does Constricts arterioles, raising total peripheral resistance
Why it matters Resistance is one of the two terms in MAP = CO × TPR, so this output moves blood pressure quickly

9 Sympathetic output to veins
What it is Norepinephrine on α_1 receptors in venous smooth muscle
What it does Venoconstriction raises venous return and stroke volume
Why it matters Pulls blood out of the reservoir when pressure falls

1 Juxtaglomerular cells (renin)
What it is Modified smooth muscle cells in the wall of the afferent arteriole of the nephron
What it does Release renin when renal pressure falls, when the macula densa sees less NaCl, or when sympathetic β_1 receptors are stimulated
Why it matters The kidney's own pressure sensor starts the cascade

7 Antidiuretic hormone (ADH, vasopressin)
What it is A peptide from the hypothalamus released by the posterior pituitary
What it does Makes the collecting duct reabsorb water; at high levels constricts arterioles
Why it matters Released by angiotensin II, by high osmolarity, and by a large fall in pressure

8 Thirst center
What it is Hypothalamic neurons that angiotensin II and high osmolarity stimulate
What it does They make you feel thirsty and drink
Why it matters Volume cannot be replaced by the kidney alone; intake has to rise

9 Atrial natriuretic peptide (ANP)
What it is A peptide released by the atria when they are stretched by too much volume
What it does Makes the kidney excrete Na^+ and water, dilates arterioles, and inhibits renin, aldosterone and ADH
Why it matters It is the brake on the whole system, released when there is already too much volume